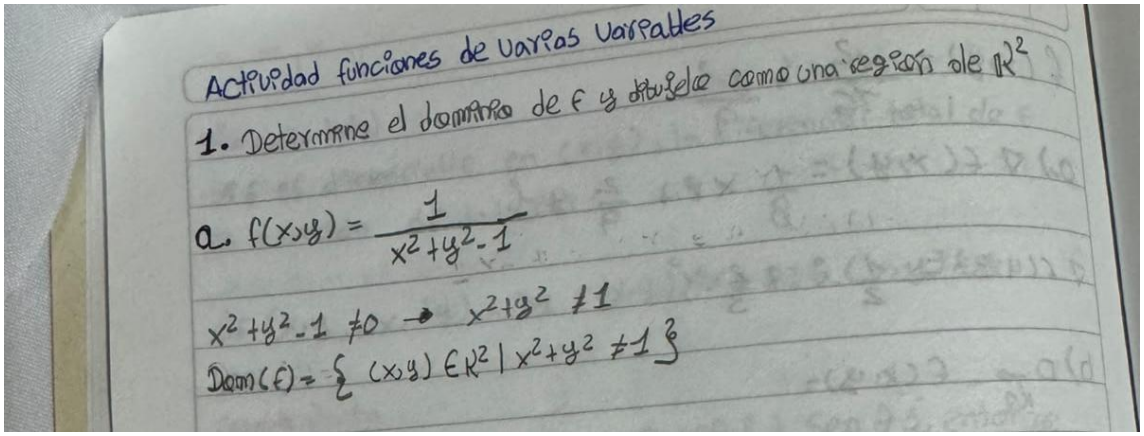
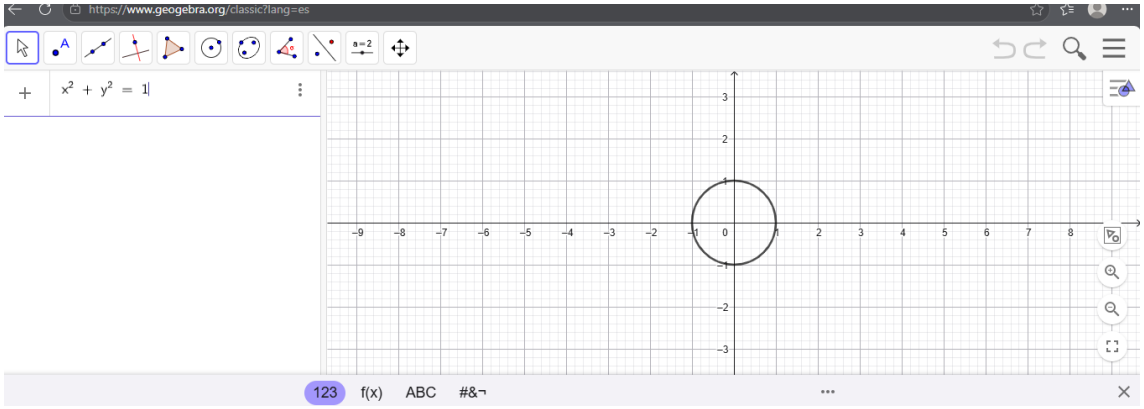
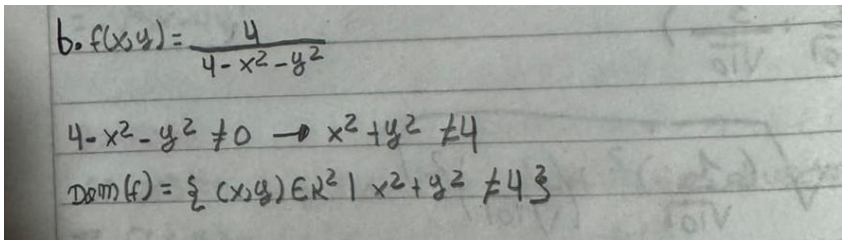
1. Determine el dominio de f y dibujelo como una region de R

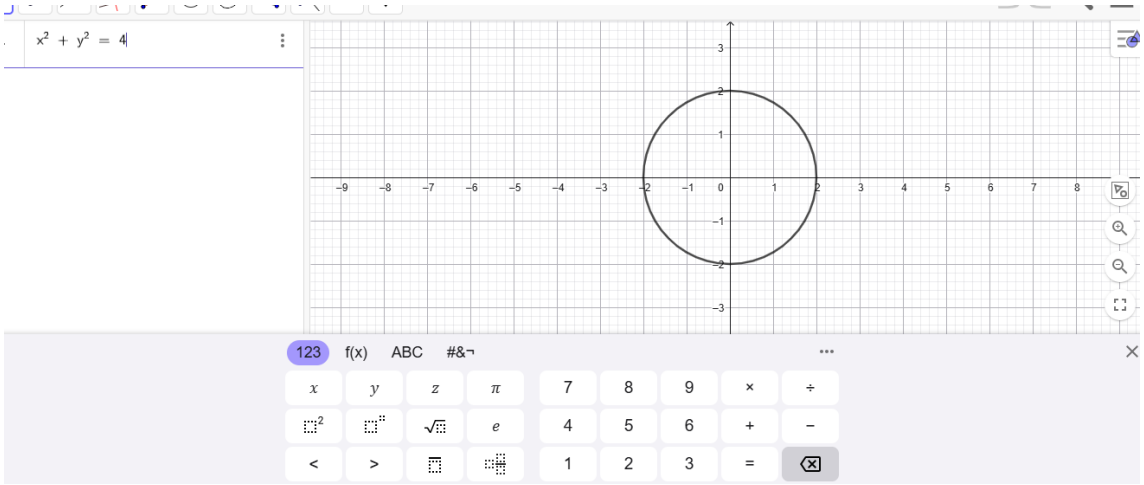
A.



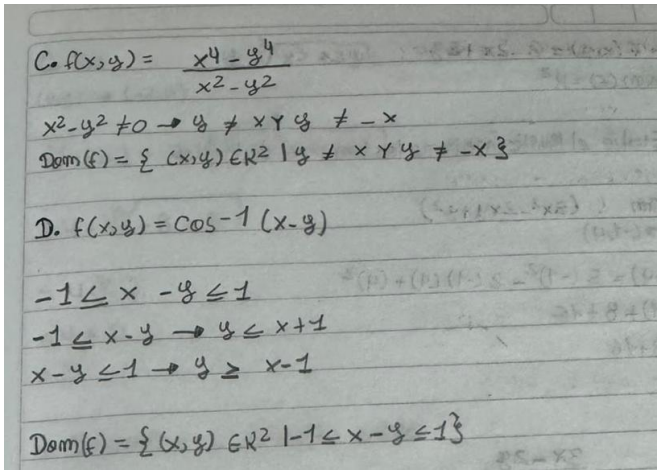


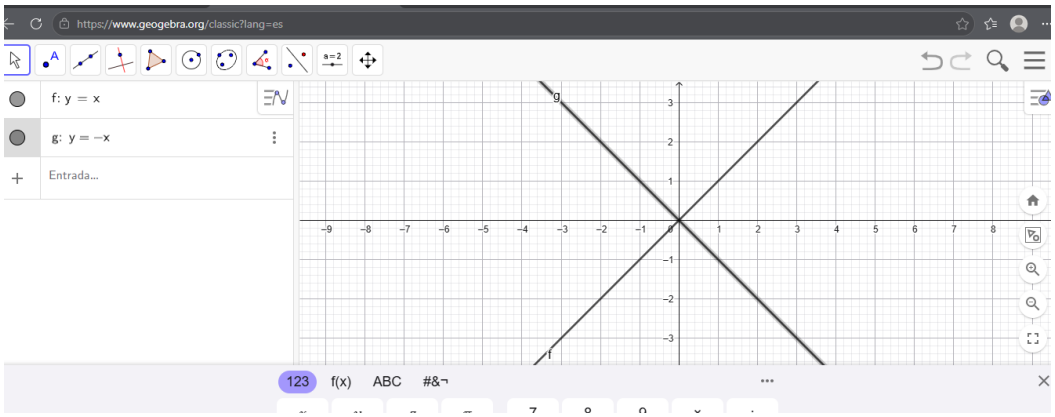
B.



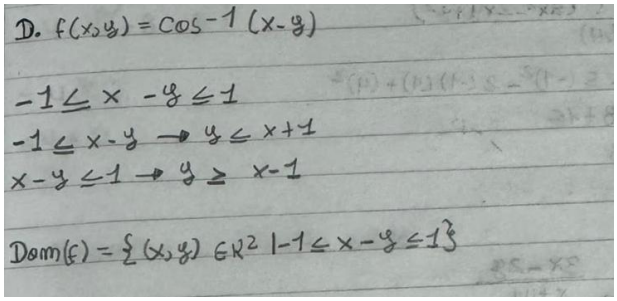


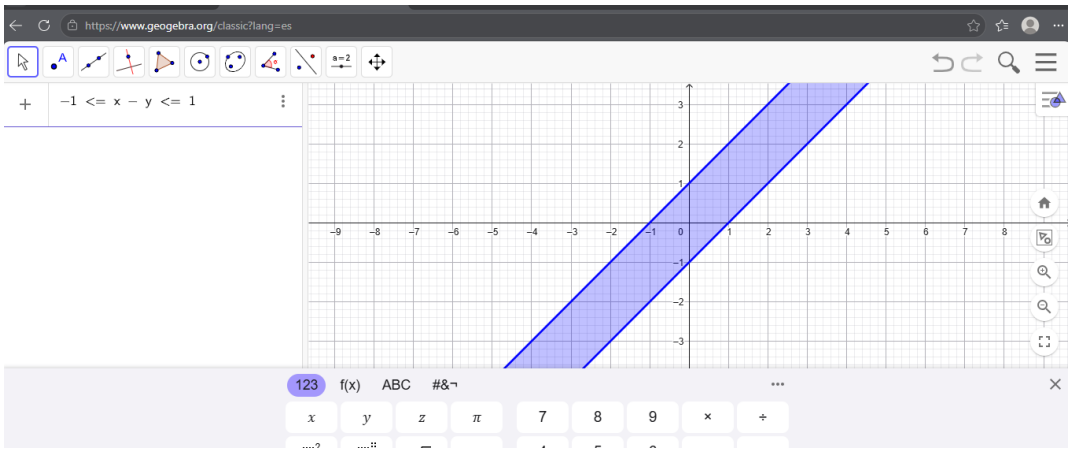
C.


C. $f(x,y) = \frac{x^4 - y^4}{x^2 - y^2}$
$x^2 - y^2 \neq 0 \rightarrow y \neq x \; y \; y \neq -x$
$Dom(f) = \{(x,y) \in \mathbb{R}^2 \mid y \neq x \; y \; y \neq -x\}$
D. $f(x,y) = \cos^{-1}(x-y)$
$-1 \leq x - y \leq 1$
$-1 \leq x - y \rightarrow y \leq x + 1$
$x - y \leq 1 \rightarrow y \geq x - 1$
$Dom(f) = \{(x,y) \in \mathbb{R}^2 \mid -1 \leq x - y \leq 1\}$


https://www.geogebra.org/classic?lang=es
f: y = x
g: y = −x
Entrada...
123
f(x)
ABC
#&¬

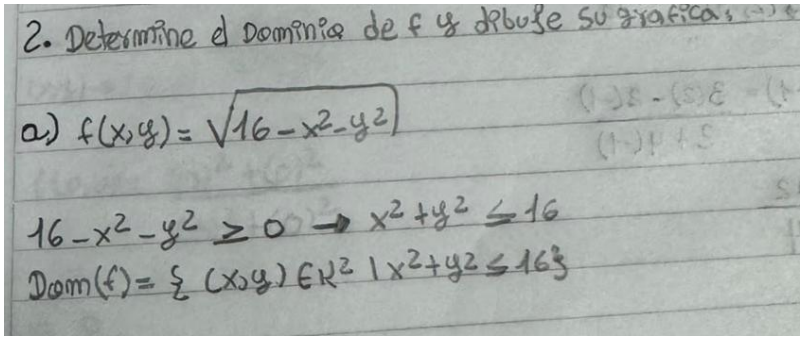
D.

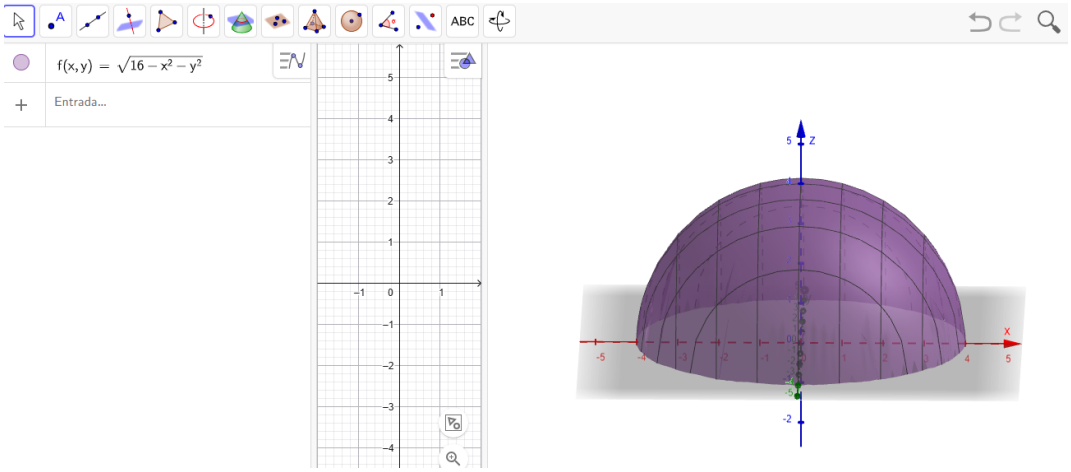

D. $f(x,y) = \cos^{-1}(x-y)$
$-1 \leq x - y \leq 1$
$-1 \leq x - y \rightarrow y \leq x + 1$
$x - y \leq 1 \rightarrow y \geq x - 1$
$Dom(f) = \{(x,y) \in \mathbb{R}^2 \mid -1 \leq x - y \leq 1\}$


https://www.geogebra.org/classic?lang=es
−1 <= x − y <= 1
123
f(x)
ABC
#&¬

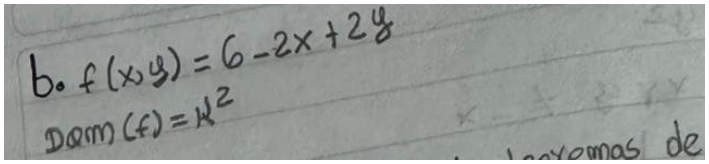
2. Determine el dominio de f y dibuje su gráfica:

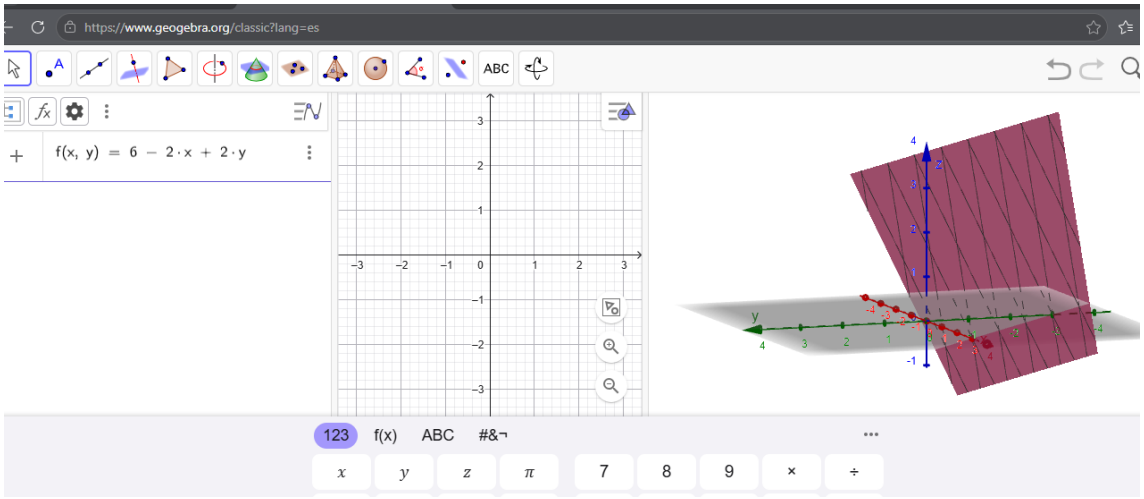
A.



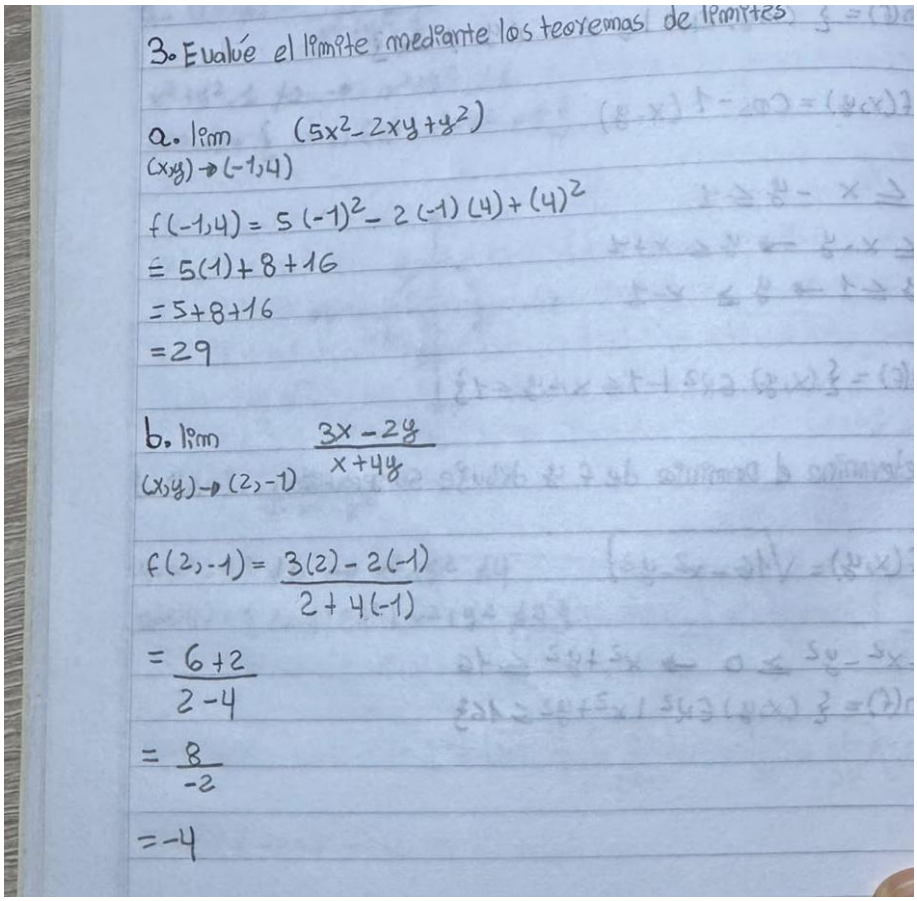


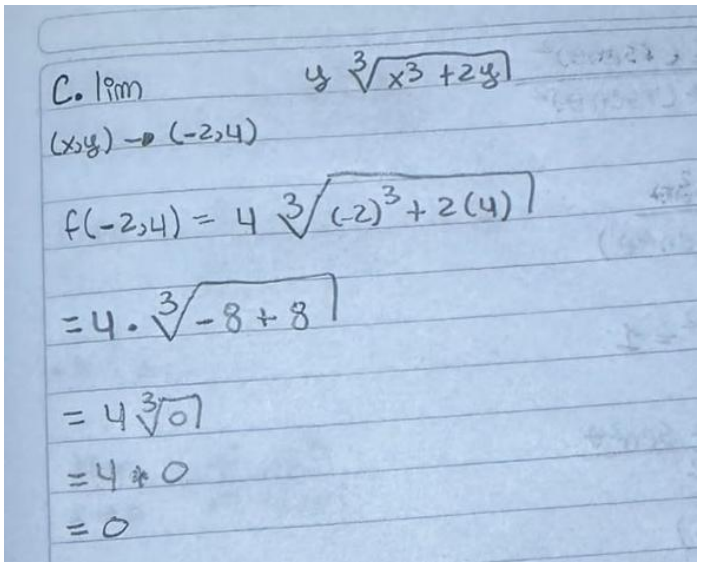
B.





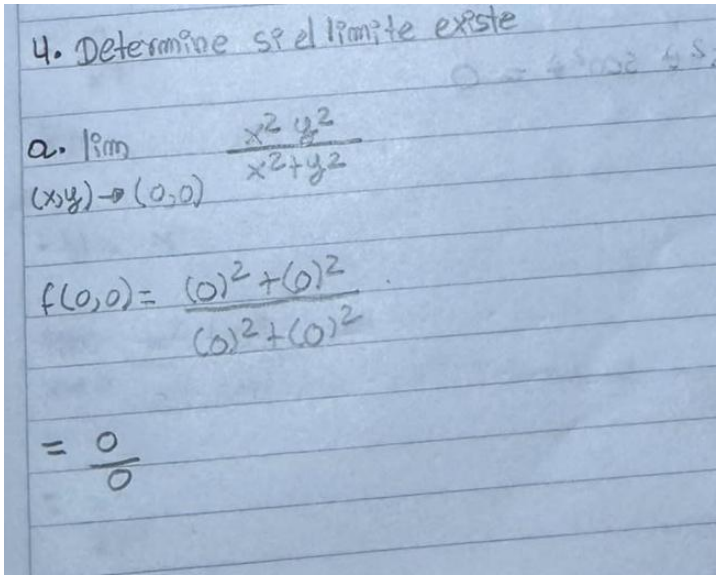
3. Evalué el límite mediante los teoremas de límites:

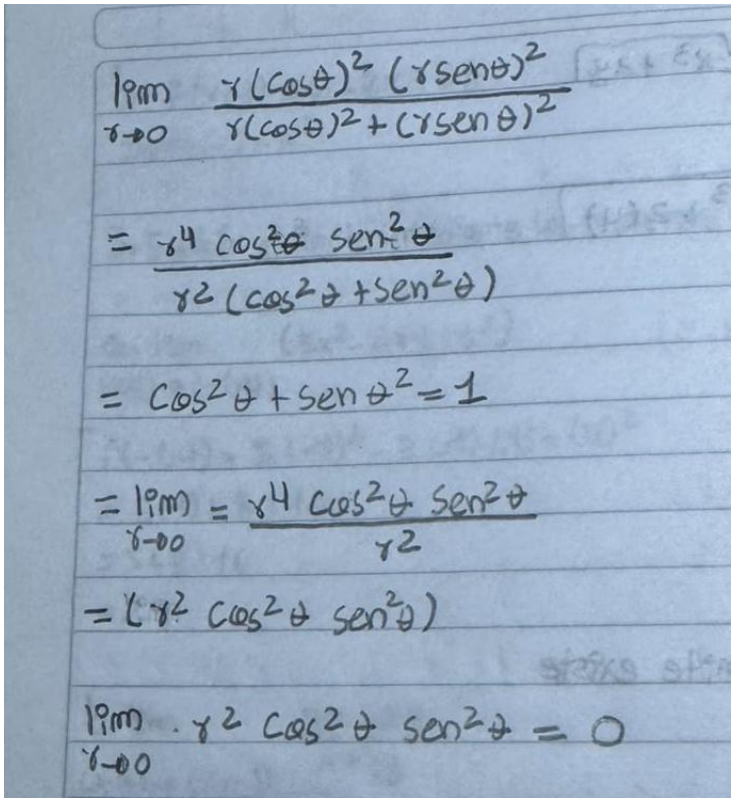




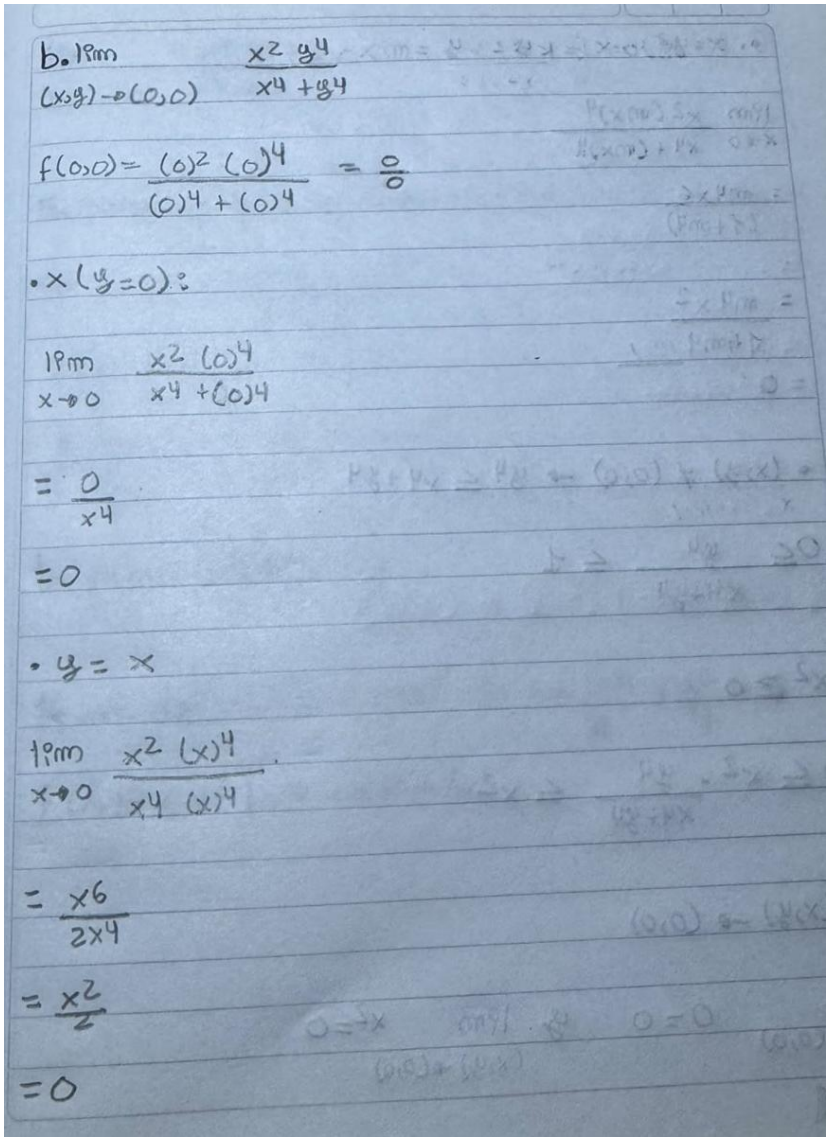
4. Determine si el límite existe:

A.





B.



b. lim $\frac{x^2 y^4}{x^4 + y^4}$
$(x,y) \to (0,0)$

$f(0,0) = \frac{(0)^2 (0)^4}{(0)^4 + (0)^4} = \frac{0}{0}$

• $x\ (y=0)$:

lim $\frac{x^2 (0)^4}{x^4 + (0)^4}$
$x \to 0$

$= \frac{0}{x^4}$

$= 0$

• $y = x$

lim $\frac{x^2 (x)^4}{x^4 (x)^4}$
$x \to 0$

$= \frac{x^6}{2x^4}$

$= \frac{x^2}{2}$

$= 0$

• $x = y^2$ o $x = ky$: $y = mx$

$$\lim_{x \to 0} \frac{x^2 (mx)^4}{x^4 + (mx)^4}$$

$$= \frac{m^4 x^6}{(1+m^4)}$$

$$= \frac{m^4 x^2}{1+m^4}$$

$$= 0$$

• $(x,y) \neq (0,0) \rightarrow y^4 \leq x^4 + y^4$

$$0 \leq \frac{y^4}{x^4+y^4} \leq 1$$

• $x^2 \geq 0$

$$0 \leq x^2 \cdot \frac{y^4}{x^4+y^4} \leq x^2$$

• $(x,y) \rightarrow (0,0)$

$\lim_{(x,y)(0,0)} 0 = 0$ y $\lim_{(x,y) \to (0,0)} x^2 = 0$

$$\lim_{(x,y)(0,0)} \frac{x^2 y^4}{x^4+y^4} = 0$$

5. Determine todos los puntos en los que la función es continua:

